

Chapter 4 — Identity, Authentication, Authority, and Delegation

Purpose and scope

Separates identity, identifiers, authentication, trust, permission, authority, delegation, policy decisions, and protected action.

Source authority

Book I: Book I Decisions 5, 6, 25, 26, 27, and 48

Book II: Book II Chapters 04, 05, 18, 20, 21, and 26

Book III: Book III Chapters 03, 05, 06, and 08

Normative semantic rules

- Identity, Authentication, Trust, Permission, Authority, Delegation, Capability, and Credential MUST remain separate semantic concepts.
- Permission MUST be represented as a contextual decision result, while Authority MUST be represented as the governed scope that constrains that decision.
- Trust MUST NOT grant Authority, and Capability MUST NOT imply Permission.
- A Delegation MUST be attributable, scoped, conditional, expiring, revocable, and bounded by the delegator's Authority.

Canonical terms

HAL-TERM-0031 — Identity

The durable governed identity by which HAL recognizes a human, device, service, sensor, agent, node, or subsystem as the same entity across time.

Distinction: Identity is distinct from names, identifiers, attributes, roles, credentials, trust, authority, and Presence.

Type/category/status: Governed entity identity / Identity / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 6, 27, and 48

Book II: Book II Chapters 04 and 05

Book III: Book III Chapter 5

Example: A dependent artifact cites `HAL-TERM-0031` when it uses **Identity** with this exact governed meaning: The durable governed identity by which HAL recognizes a human, device, service, sensor, agent, node, or subsystem as the same entity across time.

Counterexample: A dependent artifact uses **Identity** in a way that violates its required distinction: Identity is distinct from names, identifiers, attributes, roles, credentials, trust, authority, and Presence.

Relationships: HAL-REL-0007, HAL-REL-0008, HAL-REL-0009

Lifecycle: None registered

Constraint: Identity is distinct from names, identifiers, attributes, roles, credentials, trust, authority, and Presence.

HAL-TERM-0032 — Principal

An Identity that may be the subject of authentication, policy evaluation, authority, delegation, accountability, or action attribution.

Distinction: Principal status does not itself grant authority.

Type/category/status: Governed actor role / Identity / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 6, 27, and 48

Book II: Book II Chapters 04 and 05

Book III: Book III Chapter 5

Example: A dependent artifact cites `HAL-TERM-0032` when it uses **Principal** with this exact governed meaning: An Identity that may be the subject of authentication, policy evaluation, authority, delegation, accountability, or action attribution.

Counterexample: A dependent artifact uses ****Principal**** in a way that violates its required distinction: Principal status does not itself grant authority.

Relationships: None registered

Lifecycle: None registered

Constraint: Principal status does not itself grant authority.

HAL-TERM-0033 — Identity Record

The authoritative governed state representing an Identity, including its immutable identity reference, type, lifecycle, handles, relationships, and source-approved metadata.

Distinction: The record represents an Identity but is not interchangeable with the Identity.

Type/category/status: Authoritative record / Identity / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 6, 27, and 48

Book II: Book II Chapters 04 and 05

Book III: Book III Chapter 5

Example: A dependent artifact cites `HAL-TERM-0033` when it uses ****Identity Record**** with this exact governed meaning: The authoritative governed state representing an Identity, including its immutable identity reference, type, lifecycle, handles, relationships, and source-approved metadata.

Counterexample: A dependent artifact uses ****Identity Record**** in a way that violates its required distinction: The record represents an Identity but is not interchangeable with the Identity.

Relationships: HAL-REL-0007

Lifecycle: None registered

Constraint: The record represents an Identity but is not interchangeable with the Identity.

HAL-TERM-0034 — Identifier

A value used to reference an Identity or another entity within a declared namespace and lifecycle.

Distinction: Possession or presentation of an Identifier does not authenticate identity or grant authority.

Type/category/status: Reference value / Identity / Approved

Aliases: ID

Source basis: Direct source normalization

Book I: Book I Decisions 6, 27, and 48

Book II: Book II Chapters 04 and 05

Book III: Book III Chapter 5

Example: A dependent artifact cites `HAL-TERM-0034` when it uses ****Identifier**** with this exact governed meaning: A value used to reference an Identity or another entity within a declared namespace and lifecycle.

Counterexample: A dependent artifact uses ****Identifier**** in a way that violates its required distinction: Possession or presentation of an Identifier does not authenticate identity or grant authority.

Relationships: HAL-REL-0008

Lifecycle: None registered

Constraint: Possession or presentation of an Identifier does not authenticate identity or grant authority.

HAL-TERM-0035 — Identity Attribute

A property, assertion, or governed fact associated with an Identity and qualified by source, time, and confidence where applicable.

Distinction: An attribute is not the Identity and does not independently establish authentication, trust, or authority.

Type/category/status: Governed descriptive value / Identity / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 6, 27, and 48

Book II: Book II Chapters 04 and 05

Book III: Book III Chapter 5

Example: A dependent artifact cites `HAL-TERM-0035` when it uses ****Identity Attribute**** with this exact governed meaning: A property, assertion, or governed fact associated with an Identity and qualified by source, time, and confidence where applicable.

Counterexample: A dependent artifact uses ****Identity Attribute**** in a way that violates its required distinction: An attribute is not the Identity and does not independently establish authentication, trust, or authority.

Relationships: HAL-REL-0009

Lifecycle: None registered

Constraint: An attribute is not the Identity and does not independently establish authentication, trust, or authority.

HAL-TERM-0036 — Credential

A governed secret, key, token, certificate, biometric template, or other instrument used in an authentication protocol.

Distinction: Credential possession is evidence, not identity, trust, permission, or authority by itself.

Type/category/status: Authentication instrument / Security / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 6, 27, and 48

Book II: Book II Chapters 04 and 05

Book III: Book III Chapter 5

Example: A dependent artifact cites `HAL-TERM-0036` when it uses **Credential** with this exact governed meaning: A governed secret, key, token, certificate, biometric template, or other instrument used in an authentication protocol.

Counterexample: A dependent artifact uses **Credential** in a way that violates its required distinction: Credential possession is evidence, not identity, trust, permission, or authority by itself.

Relationships: None registered

Lifecycle: None registered

Constraint: Credential possession is evidence, not identity, trust, permission, or authority by itself.

HAL-TERM-0037 — Authentication

The evidence-based process and resulting confidence that a claimed Identity is presently genuine in a specified context.

Distinction: Authentication answers who or what is present; it does not answer what action is allowed.

Type/category/status: Assurance process and result / Identity / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 6, 27, and 48

Book II: Book II Chapters 04 and 05

Book III: Book III Chapter 5

Example: A dependent artifact cites `HAL-TERM-0037` when it uses **Authentication** with this exact governed meaning: The evidence-based process and resulting confidence that a claimed Identity is presently genuine in a specified context.

Counterexample: A dependent artifact uses **Authentication** in a way that violates its required distinction: Authentication answers who or what is present; it does not answer what action is allowed.

Relationships: HAL-REL-0010

Lifecycle: None registered

Constraint: Authentication answers who or what is present; it does not answer what action is allowed.

HAL-TERM-0038 — Authentication Evidence

One or more Evidence Objects used to assess whether a claimed Identity is presently genuine.

Distinction: Authentication Evidence informs assurance but does not itself grant authority.

Type/category/status: Evidence role / Evidence / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 6, 27, and 48

Book II: Book II Chapters 04 and 05

Book III: Book III Chapter 5

Example: A dependent artifact cites `HAL-TERM-0038` when it uses **Authentication Evidence** with this exact governed meaning: One or more Evidence Objects used to assess whether a claimed Identity is presently genuine.

Counterexample: A dependent artifact uses **Authentication Evidence** in a way that violates its required distinction: Authentication Evidence informs assurance but does not itself grant authority.

Relationships: HAL-REL-0010

Lifecycle: None registered

Constraint: Authentication Evidence informs assurance but does not itself grant authority.

HAL-TERM-0039 — Trust

Multidimensional, domain-specific confidence derived from evidence about an entity, claim, process, or relationship.

Distinction: Trust may inform decisions but must not be treated as authority or permission.

Type/category/status: Evidence-based assessment / Trust / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 5, 25, 26, 27, and 48

Book II: Book II Chapters 03, 05, and 18

Book III: Book III Chapters 3, 5, and 8

Example: A dependent artifact cites `HAL-TERM-0039` when it uses **Trust** with this exact governed meaning:

Multidimensional, domain-specific confidence derived from evidence about an entity, claim, process, or relationship.

Counterexample: A dependent artifact uses **Trust** in a way that violates its required distinction: Trust may inform decisions but must not be treated as authority or permission.

Relationships: HAL-REL-0014

Lifecycle: None registered

Constraint: Trust may inform decisions but must not be treated as authority or permission.

HAL-TERM-0040 — Permission

A scoped result of authoritative policy evaluation allowing a specified Principal to attempt a specified operation under stated context and conditions.

Distinction: Permission is not identical to Authority, Delegation, Capability, role, trust, or credential possession.

Type/category/status: Decision result / Authority / Approved

Aliases: authorization result

Source basis: Direct source normalization

Book I: Book I Decisions 5, 25, 26, 27, and 48

Book II: Book II Chapters 03, 05, and 18

Book III: Book III Chapters 3, 5, and 8

Example: A dependent artifact cites `HAL-TERM-0040` when it uses **Permission** with this exact governed meaning: A scoped result of authoritative policy evaluation allowing a specified Principal to attempt a specified operation under stated context and conditions.

Counterexample: A dependent artifact uses **Permission** in a way that violates its required distinction: Permission is not identical to Authority, Delegation, Capability, role, trust, or credential possession.

Relationships: HAL-REL-0012, HAL-REL-0013

Lifecycle: None registered

Constraint: Permission is not identical to Authority, Delegation, Capability, role, trust, or credential possession.

HAL-TERM-0041 — Authority

The constitutionally and policy-governed scope within which a Principal may decide or cause action.

Distinction: Authority constrains Permission evaluation but is not itself a Permission result and is not inferred from identity, trust, capability, role, proximity, or credentials.

Type/category/status: Governed decision and action scope / Authority / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 5, 25, 26, 27, and 48

Book II: Book II Chapters 03, 05, and 18

Book III: Book III Chapters 3, 5, and 8

Example: A dependent artifact cites `HAL-TERM-0041` when it uses **Authority** with this exact governed meaning: The constitutionally and policy-governed scope within which a Principal may decide or cause action.

Counterexample: A dependent artifact uses **Authority** in a way that violates its required distinction: Authority constrains Permission evaluation but is not itself a Permission result and is not inferred from identity, trust, capability, role, proximity, or credentials.

Relationships: HAL-REL-0011, HAL-REL-0013

Lifecycle: None registered

Constraint: Authority constrains Permission evaluation but is not itself a Permission result and is not inferred from identity, trust, capability, role, proximity, or credentials.

HAL-TERM-0042 — Delegation

An attributable, scoped, conditional, expiring, and revocable grant of Authority from an authorized delegator to a recipient.

Distinction: A Delegation cannot grant authority the delegator does not possess or make a constitutional invariant waivable.

Type/category/status: Governed authority grant / Authority / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 5, 25, 26, 27, and 48

Book II: Book II Chapters 03, 05, and 18

Book III: Book III Chapters 3, 5, and 8

Example: A dependent artifact cites `HAL-TERM-0042` when it uses **Delegation** with this exact governed meaning: An attributable, scoped, conditional, expiring, and revocable grant of Authority from an authorized delegator to a recipient.

Counterexample: A dependent artifact uses **Delegation** in a way that violates its required distinction: A Delegation cannot grant authority the delegator does not possess or make a constitutional invariant waivable.

Relationships: HAL-REL-0011

Lifecycle: HAL-TRANS-0005, HAL-TRANS-0006, HAL-TRANS-0007

Constraint: A Delegation cannot grant authority the delegator does not possess or make a constitutional invariant waivable.

HAL-TERM-0043 — Policy

A governed set of rules evaluated against identity, authority, delegation, purpose, context, resource, and risk inputs.

Distinction: A Policy is not itself a decision and cannot outrank its Normative Source.

Type/category/status: Decision rule set / Authority / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 5, 25, 26, 27, and 48

Book II: Book II Chapters 03, 05, and 18

Book III: Book III Chapters 3, 5, and 8

Example: A dependent artifact cites `HAL-TERM-0043` when it uses ****Policy**** with this exact governed meaning: A governed set of rules evaluated against identity, authority, delegation, purpose, context, resource, and risk inputs.

Counterexample: A dependent artifact uses ****Policy**** in a way that violates its required distinction: A Policy is not itself a decision and cannot outrank its Normative Source.

Relationships: HAL-REL-0012

Lifecycle: None registered

Constraint: A Policy is not itself a decision and cannot outrank its Normative Source.

HAL-TERM-0044 — Policy Decision Record

The attributable record of a policy evaluation, including inputs, governing policy version, outcome, obligations, reason, and correlation data.

Distinction: It records a decision; it does not create standing Authority beyond that decision's scope.

Type/category/status: Decision record / Authority / Approved

Aliases: authorization decision record

Source basis: Direct source normalization

Book I: Book I Decisions 5, 25, 26, 27, and 48

Book II: Book II Chapters 03, 05, and 18

Book III: Book III Chapters 3, 5, and 8

Example: A dependent artifact cites `HAL-TERM-0044` when it uses ****Policy Decision Record**** with this exact governed meaning: The attributable record of a policy evaluation, including inputs, governing policy version, outcome, obligations, reason, and correlation data.

Counterexample: A dependent artifact uses ****Policy Decision Record**** in a way that violates its required distinction: It records a decision; it does not create standing Authority beyond that decision's scope.

Relationships: HAL-REL-0014

Lifecycle: None registered

Constraint: It records a decision; it does not create standing Authority beyond that decision's scope.

HAL-TERM-0045 — Protected Action

An Action whose authority, privacy, security, safety, irreversibility, or constitutional impact requires explicit governed evaluation and evidence.

Distinction: A routine implementation detail is not protected merely because it is technically complex.

Type/category/status: Risk classification / Authority / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 5, 25, 26, 27, and 48

Book II: Book II Chapters 03, 05, and 18

Book III: Book III Chapters 3, 5, and 8

Example: A dependent artifact cites `HAL-TERM-0045` when it uses ****Protected Action**** with this exact governed meaning: An Action whose authority, privacy, security, safety, irreversibility, or constitutional impact requires explicit governed evaluation and evidence.

Counterexample: A dependent artifact uses ****Protected Action**** in a way that violates its required distinction: A routine implementation detail is not protected merely because it is technically complex.

Relationships: HAL-REL-0052

Lifecycle: None registered

Constraint: A routine implementation detail is not protected merely because it is technically complex.

HAL-TERM-0164 — Owner Authorization Ceremony

The Book II-governed mechanism through which the Owner authorizes an exact protected change, capability-class decision, Treaty, or other Owner-reserved matter bound to an immutable decision identifier, declared scope, and validity

period.

Distinction: It is not conversational consent, ordinary Delegation, reusable standing permission, or a substitute for Constitutional Kernel validation; it authorizes only the exact recorded matter.

Type/category/status: Protected authorization mechanism / Authority / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 48, 49, 50, and 58

Book II: Book II Chapters 03, 05, 16, 21, and 29

Book III: Book III Chapters 1, 5, 7, 8, and 9

Example: The Owner authorizes Treaty `TRT-2048` through a ceremony record bound to that exact immutable Treaty digest, scope, activation window, and decision identifier.

Counterexample: A chat message saying “I approve future treaties with this partner” is treated as a reusable Owner authorization.

Relationships: HAL-REL-0051, HAL-REL-0052

Lifecycle: None registered

Constraint: It is not conversational consent, ordinary Delegation, reusable standing permission, or a substitute for Constitutional Kernel validation; it authorizes only the exact recorded matter.

Relationship and lifecycle rules

Relationships involving these terms **MUST** use the typed relationship register. Lifecycle-bearing concepts **MUST** use the governed transition register; a status label alone never authorizes a prohibited transition.

Term-specific semantic evidence

Every Term Record above contains a governed example, counterexample, constraint, relationship reference set, and lifecycle reference set. “None registered” is explicit.

Anti-patterns

Semantic drift, authority laundering, and collapsing an entity with its identifier, record, attributes, role, or state are prohibited.

Verification

Verify ID and label uniqueness, precise source traceability, source-basis classification, relationship and lifecycle consistency, ambiguity controls, machine-readable parity, and cross-format content parity.

Change and deprecation

Every semantic change requires authority, compatibility, dependency impact, migration, reviewer, effective version, and any replacement and sunset conditions.

Review findings

Constitutional, architecture, engineering, semantic, usability, parity, and Owner-threshold reviews passed. No Owner Review item.